LAB 06 Classes

October 26, 2026

[ ]:
```python
# We first import useful packages
import numpy as np
from random import seed
from random import random
```

1 Classes, and their Objects

One very powerful feature of advanced high level programming languages is the ability to define Objects, which combine data and functions/methods. In order to create an Object, we first define a Class (the 'type' of the new object), and then Objects becoe instances of this Class.

For example a Class of type Car could be defined as:

```
class Car():
"""Model a Car."""
def __init__(self, make, model):
    """Initialise"""    self.make = make
    self.model = model
    self.tank_cap = 60          #60 litre tank
    self.tank = 0               #initial fuel fill
def fillup(self):
    """fill tank to brim"""
    self.tank = self.tank_cap
    print("Full!")

def ...                          #other functions and private vars```
and then we could use that Class to create a specific Object, and then operate on that Object
```Python
myCar = Car('nissan', 'micra')   # create/instantiate an object called myCar

print(myCar.make)                # do stuff with my new object
print(myCar.model)
myCar.fillup()
```

2 Exercise - roguelike Room Objects (5 marks)

Last time you created some helper functions to draw roguelike rooms. Now we will create a Class called a Room, from which we can make room objects.
```

- define a class of type Room, with an `__init__` function which takes two values, a width and a height, and uses those to define the object's room size. You should also define class variables `x`,`y` which give the initial location of the user in the room (best to make this location (0,0) to start with).
- write a helper function/method for your class which allows the room to draw itself
- write class functions/methods named `left`, `right`, `up`, `down` which change he location of the user in the room - taking proper account of the room walls!
- Your main program should then be the following to allow keyboard control of the user in the room:

"'Python myRoom = Room(4,4) myRoom.draw()

while True: s = input() if s=='a': myRoom.left() if s=='s': myRoom.right() if s=='q': myRoom.up() if s=='z': myRoom.down() if s=='x': print("all done") break myRoom.draw()"'

What else can your Class be programmed to do?

(Oh the `IPython.display` library has a function called `clear_output` which will clear the old output so that you can overlay the new output and give the impression of movement)

[ ]:
```
# roguelike Room Objects - write your code here



myRoom = Room(4,4)
myRoom.draw()

while True:
    s = input()
    if s=='a': myRoom.left()
    if s=='s': myRoom.right()
    if s=='q': myRoom.up()
    if s=='z': myRoom.down()
    if s=='x':
        print("all done")
        break
    myRoom.draw()
```